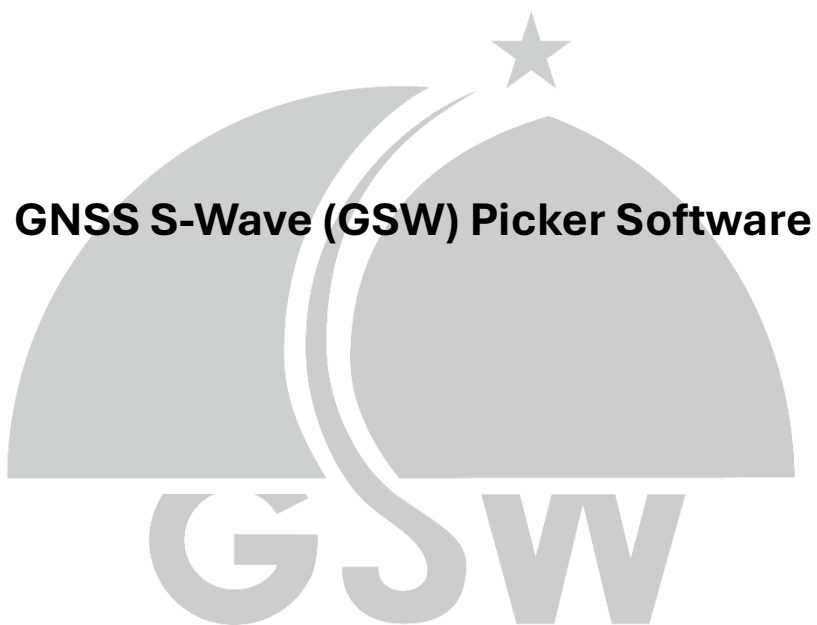


User Manual



V1.0, 2025

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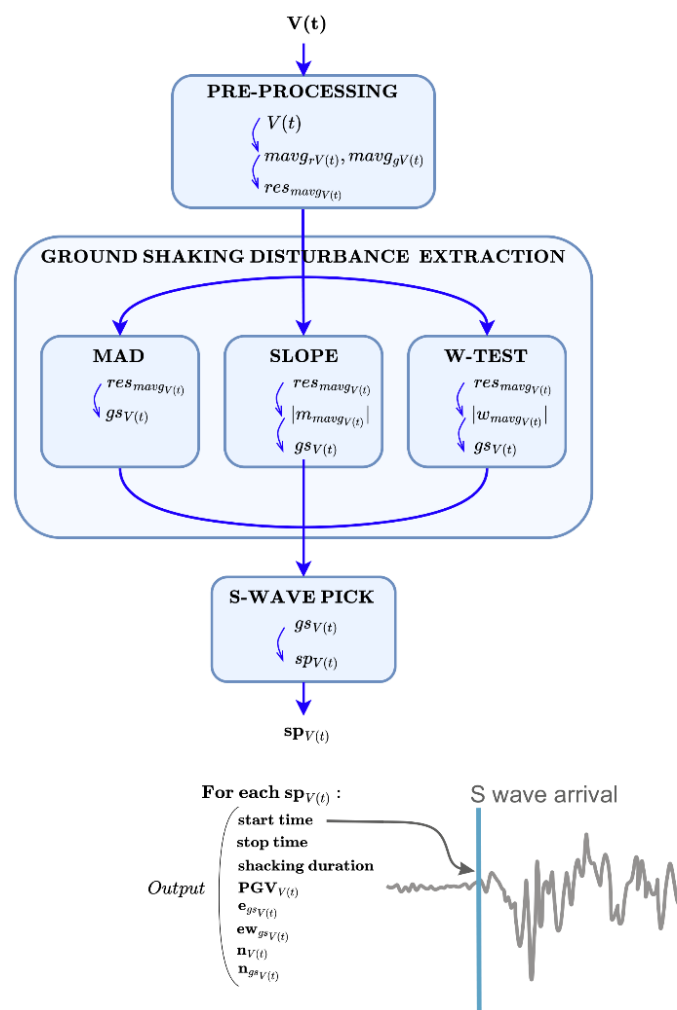
Brief Description

GSW Picker is an open-source, Python-based tool for analyzing high-rate (≥ 1 Hz) GNSS velocity data to extract **S-wave arrival times** and **ground shaking parameters** such as amplitude and duration. Designed for large-scale, research-grade seismo-geodetic applications, the software leverages asynchronous parallel processing (`concurrent.futures`) for fast and scalable performance.

The core algorithm uses a sliding-window approach to detect the **first time of arrival (ToA)** of S-waves in high-rate GNSS instantaneous velocity time series—**without requiring frequency-domain pre-filtering**.

The detection process follows three main phases (illustrated in the figure below):

- 1) **Preprocessing** – Reduces high-frequency noise in raw GNSS velocity data.
- 2) **Disturbance Detection** – Uses statistical analysis and hypothesis testing (via MAD, SLOPE, and W-TEST methods) to identify ground shaking disturbance envelopes.
- 3) **S-wave Picking** – Detects the main shock and accurately picks S-wave arrival times.



Software install

GSW Picker is compatible with Windows, macOS, and Linux (tested on all three platforms). Two installation options are provided to simplify setup on your local machine.

Start by cloning the repository to your local machine:

```
>> git clone git@github.com:HJpierzchala/GSWPicker.git
```

This will create a directory with the software files (directory structure shown in the next section).

To complete the installation one of the following setup methods can be performed:

a) Automatic Installation (Recommended)

Use the install.sh script for a guided setup.

- Detects your operating system (Windows/macOS/Linux)
- Creates a Conda environment named gsw_env with Python 3.11
- Installs all required dependencies from requirements.txt

Run the script from the root directory:

```
>> bash install.sh
```

b) Manual Installation

Use this option if the automatic method fails.

- Create conda environment

```
>> conda create -n gsw_env python = 3.11
```

- Activate the environment

```
>> conda activate gsw_env
```

- Install pip (if not already installed)

```
>> conda install pip
```

- Install required packages:

```
>> pip install -r requirements.txt
```

Directory and Project structure

GSWPicker is organized as a modular Python package. The project structure is designed for clarity and scalability, with dedicated folders for input data, results, configuration, and source code.

Directory Overview

GSWPicker/	
├─ modules/	<i># Core processing modules</i>
└─ classes.py	<i># Initializes processing per station and MODE (MAD, SLOPE, W-TEST)</i>
└─ detect.py	<i># Core module for S-wave picking</i>
└─ output.py	<i># Functions for generating tabular (.csv) results</i>
└─ parallel.py	<i># Enables parallel processing via concurrent.futures</i>
└─ prep.py	<i># Handles input data, filtering, and coordinate transformations</i>
└─ time.py	<i># Time conversions (GPST, UTC, MJD) and leap second handling</i>
└─ UI.py	<i># GUI-related functions</i>
└─ preinspect.py	<i># Tools for visualizing and analyzing raw velocity noise</i>
├─ results/	<i># Output results</i>
└─ test/	<i># Example output</i>
├─ vel_data/	<i># GNSS input data</i>
└─ test/	
└─ test_kin/	
├─ config/	<i># Configuration and dependencies</i>
└─ requirements.txt	
└─ parameters.toml	
└─ leap_seconds_table.csv	
├─ docs/	<i># Documentation</i>
└─ GSWP_user_manual.pdf	
└─ GSWP_user_manual.md	
├─ GSW_GUI.py	<i># Entry script to launch the GUI</i>
├─ mainGUI.py	<i># Terminal output and visualizer setup</i>
├─ make_tree.py	<i># Creates required directory structure</i>
├─ install.sh	<i># Installation script</i>
├─ setup.py	<i># Installation helper</i>
└─ pyproject.toml	<i># Python packaging metadata & installation requirements</i>

GSWPicker supports multiple GNSS data formats:

- **VarioPy (.varout)** – Position difference time series
- **PRIDE-PPP (kin_*)** – PPP position time series
- **Custom .oy format** – User-defined position difference time series

Input data should be placed in the `vel_data/` directory, each within its own **project folder** (created by the user).

Project Structure Overview

Each project creates a corresponding folder in the `results/` directory. A typical project result layout looks like this:

```

results/
├── <Project_ID>/
│   ├── <Axis>/
│   │   ├── figures/      # Visual outputs (plots)
│   │   ├── logs/         # Log (.log) and error (.err) files
│   │   └── reports/      # Final results in .csv format

```

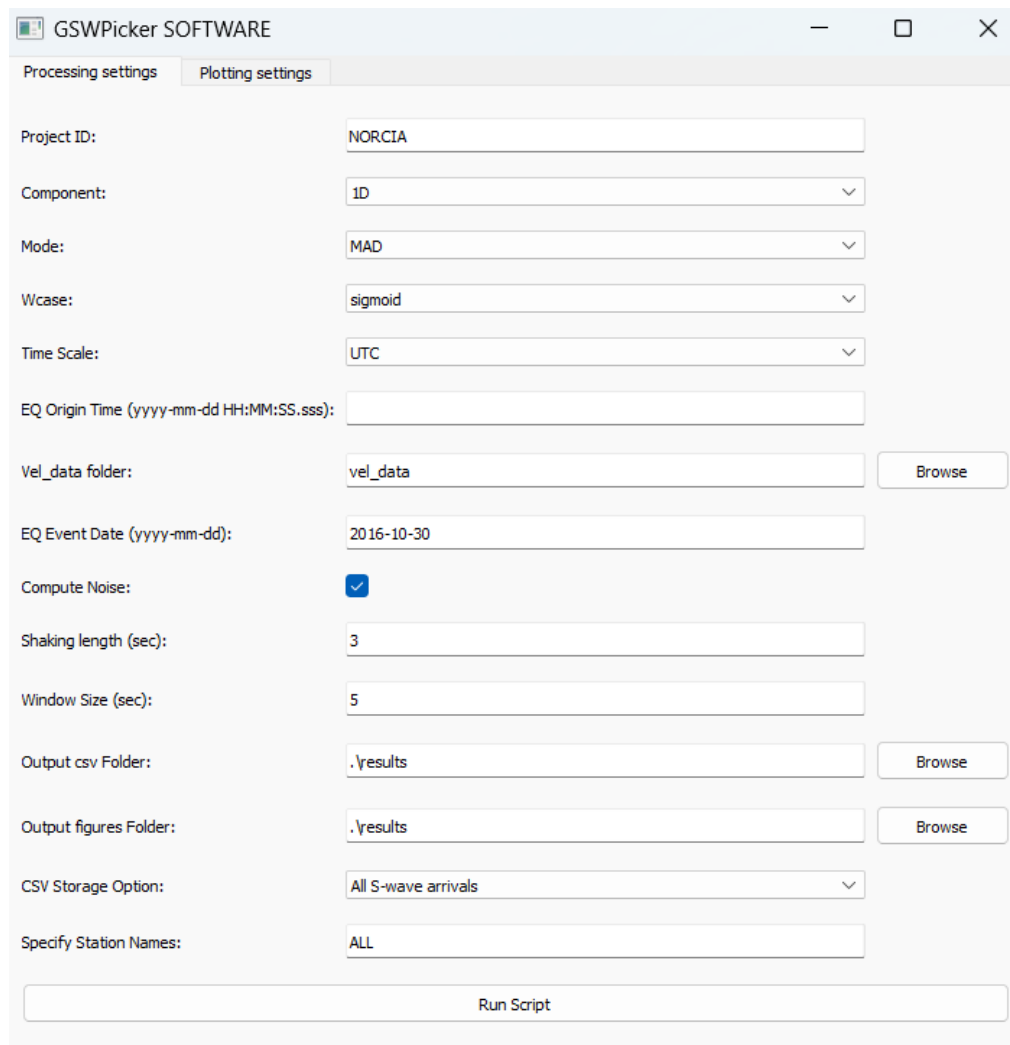
UI Parameters

GSWPicker includes a user-friendly graphical interface designed for processing large GNSS datasets with speed and flexibility. The GUI is divided into two main panels:

- Processing Settings
- Plotting Settings

Processing settings

This panel allows the user to configure the core parameters for seismic analysis.



The screenshot displays the GSWPicker SOFTWARE window with the 'Processing settings' tab selected. The settings are as follows:

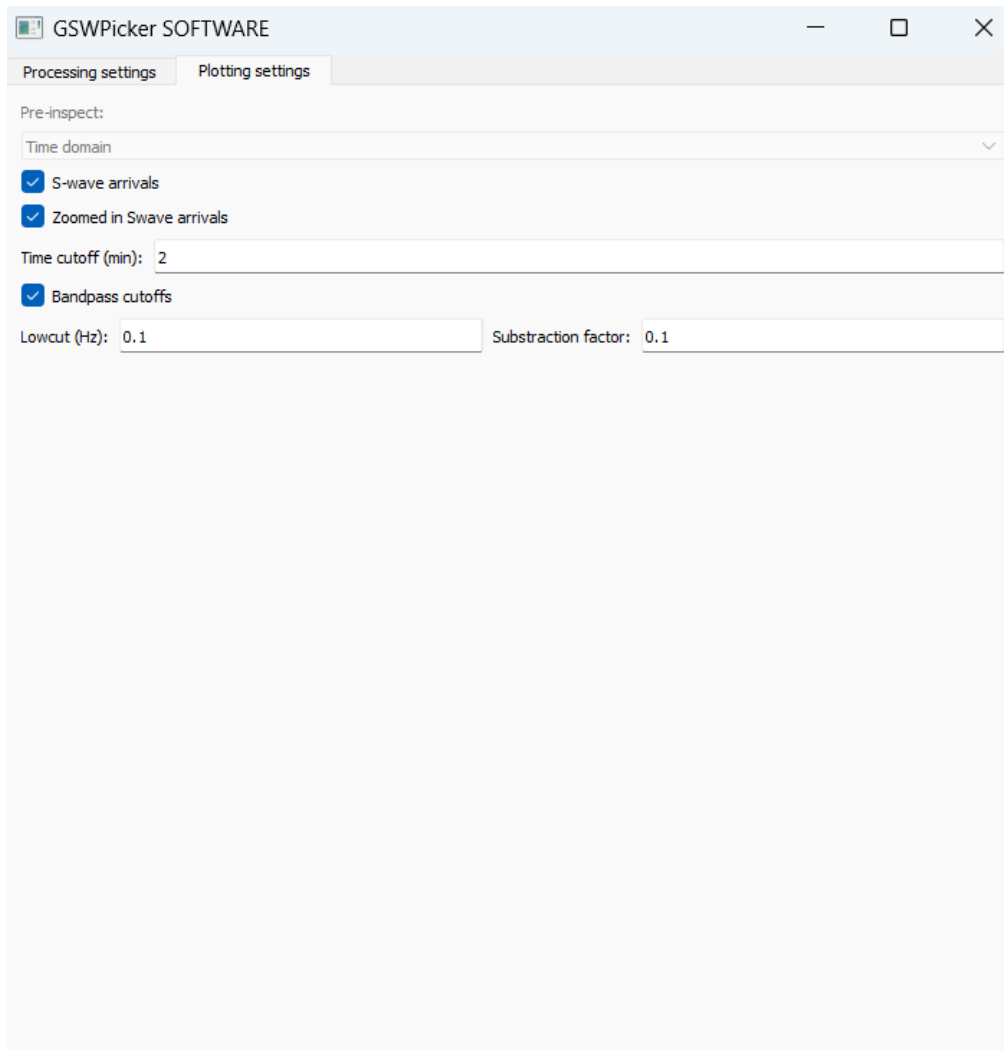
- Project ID:** NORCIA
- Component:** 1D
- Mode:** MAD
- Wcase:** sigmoid
- Time Scale:** UTC
- EQ Origin Time (yyyy-mm-dd HH:MM:SS.sss):** (empty field)
- Vel_data folder:** vel_data (with a 'Browse' button)
- EQ Event Date (yyyy-mm-dd):** 2016-10-30
- Compute Noise:** ☒
- Shaking length (sec):** 3
- Window Size (sec):** 5
- Output csv Folder:** .\results (with a 'Browse' button)
- Output figures Folder:** .\results (with a 'Browse' button)
- CSV Storage Option:** All S-wave arrivals
- Specify Station Names:** ALL

A 'Run Script' button is located at the bottom of the settings panel.

Parameter	Description
Project ID	Name of the project (used to label output directories).
Axis	Defines the spatial component(s) used for analysis: • 1D – Analyze each ENU component separately (recommended) • 2D – Combine E and N components • 3D – Combine all ENU components
Mode	Method for extracting ground shaking disturbances: • MAD (recommended) • SLOPE • W-TEST (recommended)
Wcase	Weighting scheme used with the SLOPE method. Currently only sigmoid is functional. Options: sigmoid, trapez, boxchar
Time Scale	Defines the time reference system: • UTC – Requires EQ Event Date • GSOW – GPS seconds of week • RELATIVE – Relative to earthquake origin time (requires EQ Origin Time)
EQ Origin Time	Earthquake origin time (for RELATIVE time scale). Format: YYYY-MM-DD HH:MM:SS.sss
EQ Event Date	Date of the earthquake event (required for UTC time scale). Format: YYYY-MM-DD
Vel_data Folder	Path to the vel_data directory containing input GNSS data.
Compute Noise	Checkbox to compute station noise automatically (currently always enabled unless using .oy input format).
Shaking Length (sec)	Minimum duration that defines a ground shaking period. <i>Recommended: 3 sec</i>
Window Size (sec)	Window size for moving average smoothing during pre-inspection. <i>Recommended: 5 sec</i>
Output CSV Folder	Path where the tabular output files (.csv) will be stored.
Output Figures Folder	Path for storing plots and visual outputs.
CSV Storage Option	Defines the format for saving .csv outputs: • Station dependent • All S-wave arrivals • Component-wise S-wave arrivals
Specify Station Names	Optional: list station IDs to be processed (e.g., ACCU, AMAT). Leave empty or use ALL to process all available stations.

Plotting settings

This panel allows users to configure visual output.



Parameter	Description
Pre-inspect	Enables pre-inspection diagnostics: • Time-domain – Visualizes raw velocity time series • Frequency-domain – Displays spectrograms • Noise analysis – Shows autocorrelation and Lomb-Scargle PSD of noise of velocity time series
S-wave Arrivals	Enables plotting of detected S-wave arrival times across the entire time series.
Zoomed-in S-wave Arrivals	Adds a close-up view around the picked S-wave arrival: • Time Cutoff (min) – Duration before/after the shock for zoomed view • Bandpass Cutoffs – Enable bandpass filtering for comparison view: • Lowcut (Hz) – Lower frequency limit • Subtraction Factor – Value subtracted from Nyquist frequency ($f_s/2$) to avoid filter edge issues

Output

GSWPicker generates both **tabular (.csv)** and **graphical (.png/.svg)** output based on your selections in the GUI—specifically under the parameters:

“Storage Option,”
and “S-wave arrivals,” “Zoomed-in S-wave arrivals,” “Bandpass cutoffs.”

Tabular Output (.csv)

The tabular output includes detailed metrics for each detected earthquake shock period. Each .csv file may contain the following columns:

Field	Description
S-wave detected at (sec)	Start time of the detected S-wave arrival
End of shaking at (sec)	End time of the detected shock period
Shaking duration (sec)	Duration of the detected shaking
Peak Ground Velocity (m/s)	Component-wise PGV calculated from raw velocity data
Integrated energy (m)	Total shaking energy over the shock period
Weighted energy (m)	Total energy weighted by the PGW
Sampling frequency (Hz)	Sampling rate of the GNSS velocity data
Velocity noise (m/s)	Noise level in the raw instantaneous velocity
Ground shaking noise (m/s²)	Noise level of the ground shaking disturbance envelope

CSV Output Types

Depending on the selected **CSV Storage Option**, three types of .csv outputs are generated:

1. Station Dependent

Generates a .csv file per station and component. Each file contains all detected shock periods, sorted in ascending order of "Weighted energy" (main shock appears first).

Filename format:

<Project_ID>_<Mode>_<Axis>_<Component>_<StationID>.csv

A	B	C	D	E	F	G	H	I	J	K	L
Station	component	S-wave detected at (sec)	End of shaking at (sec)	Shaking duration (sec)	Peak Ground Velocity (m/s)	Integrated energy (m)	Weighted energy (m)	Sampling frequency (Hz)	Velocity noise (m/s)	Ground shaking noise (m/s ²)	
0 ACCU	e	24042.1	24113.2	71.1	0.41	0.033881847	0.071635698	10	0.003719544	2.61E-05	
1 ACCU	e	22118	22140.1	22.1	0.2	0.004111022	0.013085046	10	0.003719544	2.61E-05	
2 ACCU	e	24129.7	24149.9	20.2	0.04	0.002194587	0.005036947	10	0.003719544	2.61E-05	
3 ACCU	e	24117.7	24123.9	6.2	0.05	0.000822013	0.002749429	10	0.003719544	2.61E-05	
4 ACCU	e	24197.6	24204.4	6.8	0.03	0.000704941	0.00225502	10	0.003719544	2.61E-05	
5 ACCU	e	24186.3	24193.1	6.8	0.03	0.000704308	0.00223351	10	0.003719544	2.61E-05	
6 ACCU	e	24295	24301.3	6.3	0.04	0.000671031	0.002216209	10	0.003719544	2.61E-05	
7 ACCU	e	23879.5	23883	3.5	0.04	0.00035263	0.001412585	10	0.003719544	2.61E-05	

2. All S-wave Arrivals

Creates a single .csv file that includes all S-wave arrivals across all stations and all components, sorted per station block (a sort within the 3 components of each station) in ascending order of “S-wave detected at (sec)”

Filename format:

<Project_ID>_<Mode>_<Axis>_ALL.csv

A	B	C	D	E
Station	component	S-wave detected at (sec)		
ACCU	n	24040.2		
ACCU	u	24041.4		
ACCU	e	24042.1		
ATTE	e	24055.7		
ATTE	n	24070.4		
ATTE	u	25145.4		
LTNA	n	24066.6		
LTNA	e	24084.2		
LTNA	u	24099.9		

3. Component-wise S-wave Arrivals

Outputs a separate .csv for each component across all stations. Each file is sorted by the "S-wave detected at (sec)" column.

Filename format:

<Project_ID>_<Mode>_<Axis>_<Component>.csv

(Expect 3 files: one each for E, N, and U components)

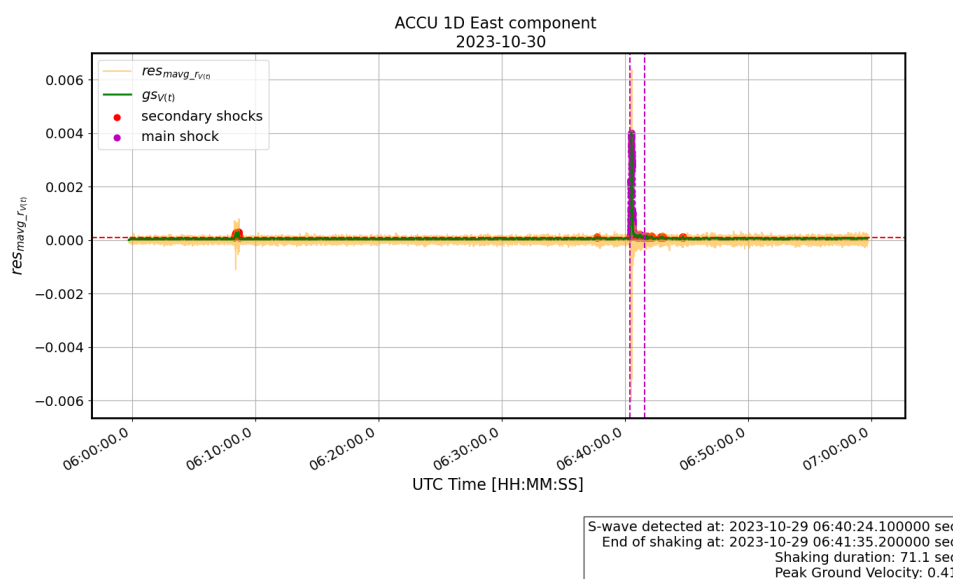
A	B	C	D	E	F	G	H	I	J	K	L	M
Station	component	S-wave detected at (sec)	End of shaking at (sec)	Shaking duration (sec)	Peak Ground Velocity	Integrated energy	Weighted energy	Sampling frequency	Velocity noise (m/s)	Ground shaking noise (m/s ²)		
ACCU	e	24042.1	24113.2	71.1	0.41	0.033881847	0.033881847	10	0.003719544	2.61E-05		
ATTE	e	24055.7	24103.7	48	0.04	0.005377362	0.005377362	10	0.003248244	1.82E-05		
LTNA	e	24084.2	24153.6	69.4	0.06	0.06129067	0.06129067	10	0.003091642	3.12E-05		

Graphical Output (.png/ .svg)

Visual outputs are stored in the “Output Figures Folder” and consist of two figure types:

1. S-wave Arrivals Plot

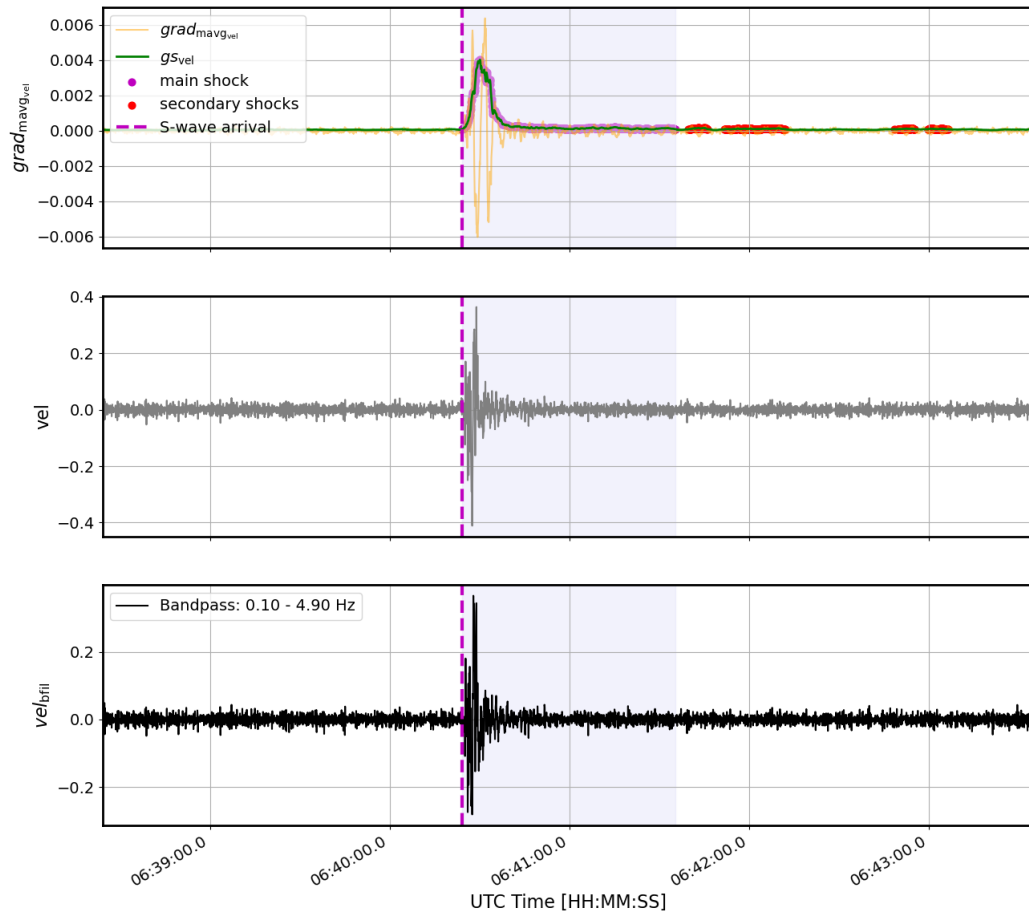
- Station-specific plot showing **all detected shocks**, including the **main shock**.
- Provides numerical values of S-wave arrival, amplitude, and duration of the shaking period.



2. Zoomed-in S-wave Arrival Plot

- **Focuses on a narrow time window** around the detected main shock.
- Compares the ground shaking duration times series with the raw velocity time series and its bandpass filtered version.
- The zoom window is defined by the user's "Time Cutoff" and "Bandpass Cutoffs" settings.

ACCU: 1D MAD East component - Comparison with original and bandpass filtered vel
2023-10-30



Thank you for using **GSWPicker**! We hope it supports your research and seismic analysis effectively.

If you have any questions, suggestions, or feedback, feel free to reach out:

- **Hubert Pierzchala** – hpierzchala@cbk.waw.pl
- **Alex Lapadat** – a.m.lapadat@tudelft.nl

We appreciate your input and are happy to help!

